Anh Dao

anhdd2000@gmail.com | +358 41 325 1677 | Portfolio

EDUCATION

M.Sc. (Technology) in Computer Science
Aalto University, Finland ***August 2024 – July 2026***
Graduated 31 July 2026. GPA: 4.09 / 5.00. Minor: Machine Learning, Data Science and Artificial Intelligence
Thesis: *Mitigating Sensor-Constrained Attacks on Virtual Reality Cybersickness Classifiers*

B.A. in Computer Science
Grinnell College, United States ***August 2019 – May 2023***
GPA: 3.76 / 4.00

Relevant Coursework: Information Security, Network Security, Platform Security, Cryptography, Cloud Software and Systems, Scalable Systems and Data Management, Programming Parallel Computers, Deep Learning, Machine Learning, Federated Learning, Machine Learning in Biomedicine, Complex Networks, Seminar on Large Language Models

EXPERIENCE

Smartly – ***Software Engineer*** ***Feb 2026 – Present***

- Working on the Agentic Orchestration team for Smartly's core initiative Synapse, across **Python**/**Pydantic AI** and **TypeScript**/**Node.js** projects.
- Building Smartly's **Pydantic AI**-based agentic orchestration harness, including progressive tool discovery, Plan Mode control flow and LLM-friendly tool interfaces, as well as Smartly's external MCP server.
- Maintaining **PostgreSQL** storage and **Redis** caching for agents' metadata, message history and draft outputs, plus PoCs for a persisted artifacts log; Redis also backs our **BullMQ** job queues.
- Unblocked 2 core teams through cross-team collaboration, shipping emergency features, bug fixes and refactors.
- Prototyped the company's 1st external MCP gateway and **TypeScript**-based MCP servers, creating the company guidelines on tool design and MCP implementation for 20+ engineering teams.
- Assisted in the 2-person migration of the company's Frontend monorepo of 4 million lines of TypeScript from **Jest** to **Vitest**, and solo-migrated 4 more repos from Jest/ESLint/Prettier to Vitest/Oxlint/Oxfmt.
- Optimized 2 existing LLM-based services through **LangFuse** trace monitoring and evals-based prompt optimization, reducing the cost of inference by 2.5x without loss of performance.
- Part of the company's AI Productivity Frontier group, delivering Knowledge Sharing sessions to 200+ R&D employees.

Musixmatch – ***Research Assistant*** ***Feb 2026 – Aug 2026***

- Built an ETL data pipeline in Python to exhaustively collect, clean, and deduplicate Musixmatch's lyrics dataset through their provisioned API, up to 10M+ tracks.
- Ran batch LLM annotation on 15,000+ Finnish tracks, cleaning up malformed LLM outputs into structured data.
- Replicated and extended prior work on Finnish hip-hop networks and rhyme analysis with cross-linguistic and literary analysis. Link.

Aalto University – ***Master's Thesis Worker*** ***Jan 2026 – Jul 2026***

- Developed feature-constrained gradient attacks and a confidence-triggered fallback defense for VR cybersickness classifiers using sensor-based LSTM models.

Nokia Solutions & Networks – ***R&D Trainee*** ***Aug 2025 – Jan 2026***

- Built a **Python** ETL pipeline indexing 100,000+ documents into **Vertex AI Search** on **GCP**, unifying product search for a RAG engine.
- Prototyped a multi-agent harness with Gemini Agent Platform combining fanout research, network logs triaging, and RAG-based QnA, reducing incident triage time by ∼40%; built the supporting LLM tools in **Python** and **React**.

Aalto University – ***Teaching Assistant*** ***Aug 2024 – 2025***

- Assisted in core Master's-level Security courses, totalling over 500 students, by holding office hours, designing network security assignments, and creating tools for fast grading.

Grinnell College (DASIL) – ***Research Assistant*** ***Aug 2022 – May 2023***

- Built multivariate statistical models in **R** and **Python** using game data from Stat2Labs, a platform used by 1,000+ students across 5 institutions.
- Helped over 150 students and faculty weekly with computing and data-related projects.

Microsec Zrt. – ***Software Engineer Trainee*** ***Jun – Aug 2022***

- Extended the company's PKI certificate linter in **Go** to handle edge cases of subordinate certificates following RFC 5280.
- Implemented parsing logic for X.509 certificate extensions and validation rules.

Grinnell College – ***Peer Educator – Computer Science Department*** ***Aug 2021 – Dec 2022***

- Hosted weekly mentor sessions for 100+ students across multiple CS courses, with coding challenges in **Java** and **C** and interactive Q&A sessions on course materials.

PROJECTS

Teleport Contest – Autonomous Coding Harness ***2026 – Present***

- Participating in the Teleport Coding Challenge, using an agent swarm to port NetHack 5.0, a complex rogue-like game with 500,000 lines of C and Lua code, to JavaScript with byte-exact behavior.
- Designing guardrails, multi-agent orchestration, evaluation mechanisms to prevent common pitfalls with independent long-horizon agents like reward hacking, challenge bypass and goal misalignment.

Personal Assistant on Cloudflare Durable Objects ***2026 – Present***

- Building a personal assistant on **Cloudflare Durable Objects** with the Pi agent harness and OptChat-based memory: a never-ending chat compressed into a summary tree, giving constant-size context the agent can zoom into.

MLX Silicon Inference assorted projects ***2026***

- Ported Open Source Decision models into an MLX-compatible framework, including Cloudflare Clef. Evaluated and optimized inference performance on an M4 Pro with relevant Jev-based evals and benchmarks.
- In separate **llama.cpp** experiments using **Metal**, speculative decoding increased Gemma 4 12B throughput by 12.1% across three tasks.

Scalable Agentic Systems Evaluation ***Jan – Apr 2026***

- Ran SWE-Bench Lite evaluations on 300 software engineering tasks, utilizing open source models like GLM-4.7, across different harnesses like mini-swe-agent and Claude Code
- Deployed **vLLM** with **ROCm** on LUMI's AMD MI250X GPUs using **SLURM**; profiled GPU utilization, memory, latency, and concurrency for 6+ open-weight LLMs on these same tasks, fixing incompatible inference formats and missing support for AMD-based inference cookbooks.

Complex Event Processing with Proper Load Shedding ***Sep – Dec 2025***

- Implemented a state-based load shedding strategy in **Python** for the **OpenCEP** framework to handle bursty workloads while maintaining low latency and acceptable recall
- Developed adaptive load shedding using weighted scoring of partial matches, with a feedback controller that adjusts the drop fraction from p90 latency and CPU/memory pressure
- Achieved 90-100% recall rates while maintaining latency limits through intelligent partial match pruning
- *Role: Semantic/hybrid load-shedding and state-management implementation. Collaborators: Quoc-Huy Trinh, Pablo Rubio*
- *Code: github.com/huyquoctrinh/Scalablesys-Assignment1*

Enhance Graph Retrieval-Augmented Generation by Caching Method ***Sep – Dec 2025***

- Enhanced a **Python**/**Kuzu** GraphRAG pipeline by integrating an LRU caching mechanism and memory-efficient caching variant combining context pruning and on-the-fly compression
- Implemented Text2Cypher component using dynamic few-shot prompting with TF-IDF-based exemplar selection for semantic similarity matching
- Achieved highest throughput and lowest latency at scale with memory-efficient LRU caching, demonstrating 60-90% memory savings for large context-heavy entries
- *Role: Few-shot prompting implementation and dataset creation. Collaborators: Quoc-Huy Trinh, Pablo Rubio*
- *Code: github.com/huyquoctrinh/ScalableSys-Proj2*

SKILLS

Languages: Python, TypeScript/JavaScript, Go, Java, C/C++, R, SQL, Cypher

Frameworks & Libraries: Pydantic AI, Node.js, React, Jotai, Express.js, Model Context Protocol (MCP), PyTorch, MLX, vLLM, LiteLLM, GEPA, DSPy, NetworkX, scikit-learn, pandas, NumPy

Tools & Platforms: Git, Docker, Kubernetes, AWS, Google Cloud Platform, Vertex AI, Cloudflare Workers, Kafka, BullMQ, SLURM, Vitest, Playwright, Oxlint, Oxfmt, LangFuse, Prometheus, Grafana, Codex, CUDA

Databases: PostgreSQL, Redis, Kuzu (Graph DB), Neo4j

Concepts: Agent Harnesses, LLM Evaluation, Prompt Optimization, Adversarial Machine Learning, Machine Learning, Deep Learning, Graph Neural Networks, Diffusion Models, RAG Systems, Complex Event Processing, Network Security, Zero-trust Architecture, PKI/X.509, Cryptography